Direct Stiffness Method Program for 3D Structural Analysis

A Simplified DSM Solver and Validation Example

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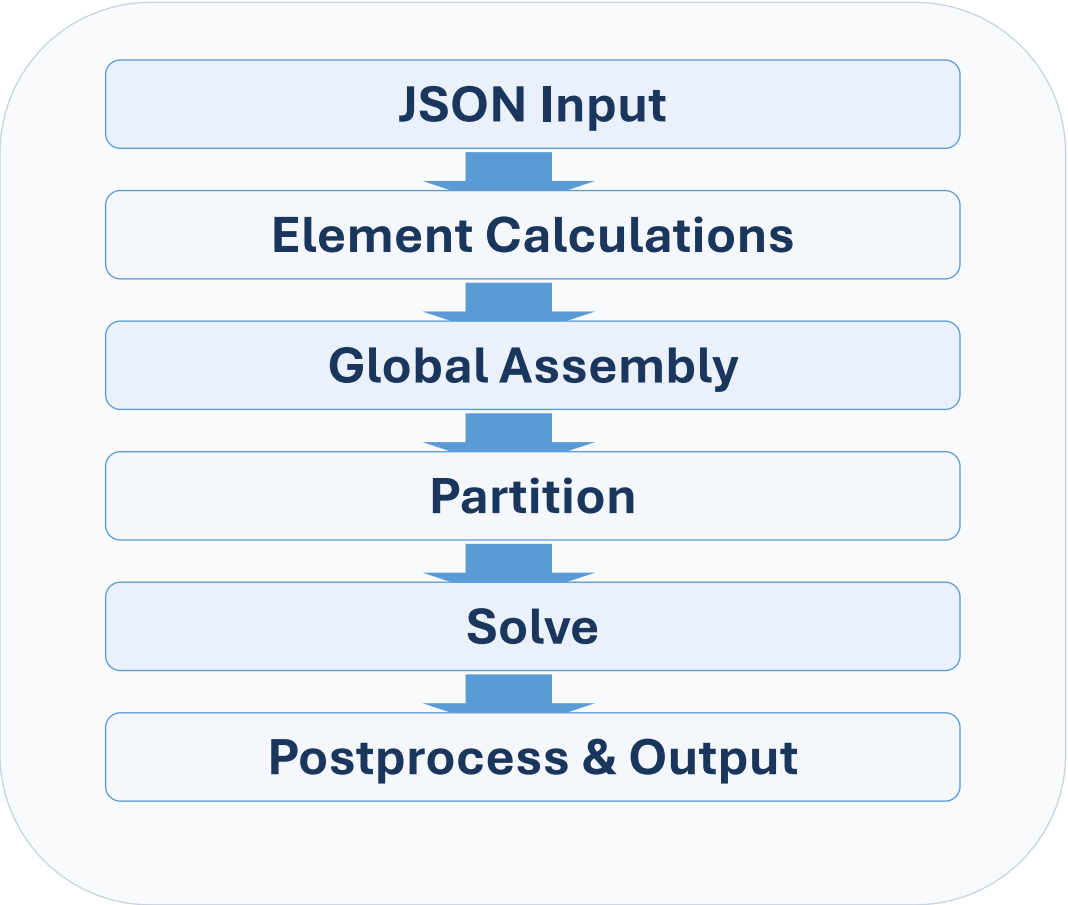
Project Objective and Workflow

Overview

- A Direct Stiffness Method program is developed for 3D truss, frame, and hybrid truss-frame systems.
- A JSON input file is used to define geometry, supports, materials, sections, and loading.
- One consistent pipeline is followed: read input → element calculations → assembly → partition → solve → postprocess.
- Displacement, reaction, and member-force tables are exported, and the deformed shape is visualized in the notebook.



Validation Strategy

How to reproduce

By running **main.ipynb** sequentially, the analysis can be reproduced.

Truss Validation (4 cases)

Nodal loads

Support displacement

Temperature load

Fabrication error

Frame Validation (6 cases)

Nodal moments

Distributed member loads

Moment release

Support displacement

Temperature load

Fabrication error

Truss Validation I: Nodal Loads and Support Displacement

Nodal Loads

Node 11: $F_z = -100\text{kN}$

Node 12: $F_x = +100\text{kN}$

- Key check: reactions balance the applied forces
- Observed behavior: reaction sums are consistent with global equilibrium

Support Displacement

Node 1: prescribed $u_z = +0.01\text{m}$

- Key check: restrained-node displacement should match the input exactly
- Observed behavior: the displacement table reports the imposed value correctly
- Interpretation: this validates boundary-condition handling

Truss Validation II: Temperature and Fabrication Error

Temperature Load

All 43 members: $\Delta T = +30°C$

Thermal coefficient: $\alpha = 1 \times 10^{-5}$

- Expected axial strain: $\alpha \Delta T = +3.0 \times 10^{-4}$
- Purpose: verify thermal-load assembly in the truss solver

Fabrication Error

Element 6: $+0.01\mathrm{m}$ length error

Element 17: $-0.01\mathrm{m}$ length error

- Purpose: convert local incompatibility into equivalent element actions
- Observed behavior: moderate redistributed displacement field

Frame Validation I: Nodal Moments and Member Loads

Nodal Moments

Selected M_x and M_y nodal moments are applied at intermediate floors

- Purpose: show directional deformation in the 3D frame model
- The case demonstrates bending-dominated frame behavior rather than pure axial response

Distributed Member Loads

12 members: $q_x = +25\text{kN/m}$

Total applied x-direction load: $2000\ \text{kN}$

- Key check: reactions sum to -$2000\ \text{kN}$ in F_x
- Purpose: verify member-load conversion and reaction recovery

Frame Validation II: Moment Release and Support Displacement

Moment Release

Use the same member-load case with selected release codes $\mathrm{MT}=1$ or 2

- Expected trend: the released structure becomes softer
- Observed result: max displacement rises from $12.764\ \mathrm{m}$ to $16.353\ \mathrm{m}$
- Displacement increase: about 28.1%

Support Displacement

Node 1: $(-0.01, -0.01, -0.01)\,\mathrm{m}$

Node 3: $(+0.01, +0.01, +0.01)\,\mathrm{m}$

- Key check: restrained-node displacements should equal the imposed values
- Purpose: verify prescribed support movement in the frame solver

Frame Validation III: Temperature and Fabrication Error

Temperature Load

All 43 members: $\Delta T = -30°C$

Thermal coefficient: $\alpha = 1 \times 10^{-5}$

- Expected axial strain: $\alpha \Delta T = -3.0 \times 10^{-4}$
- Purpose: verify thermal-load assembly in the truss solver

Fabrication Error

Element 5 and element 6: error = -0.01 m

- Purpose: convert local incompatibility into equivalent element actions
- Observed behavior: moderate redistributed displacement field

Case Study: Jiujiang Yangtze River Bridge

Three-span continuous arch-truss bridge

Total length: 576 m; span arrangement: 180 + 216 + 180 m

Modeled as a mixed truss-frame system

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Final Bridge Model and Load Cases

Model

Three-span continuous arch-truss bridge based on the Jiujiang Yangtze River Bridge configuration.

354 nodes and 1225 elements.

Mixed idealization: 1005 truss elements + 220 frame elements.

Load Files

Load case	Main loading description
Nodal loads only	130 loaded nodes with concentrated downward loads (full 3 spans)
Member loads only	350 members with $q_{\text{global}} = (0, -1, 0)$ kN/m (full 3 spans)
Support disp only	4 internal support nodes with (middle 4 supports)
Temperature only	All members: $\Delta T = +50°C$ $\alpha = 1 \times 10^{-5}$
Comprehensive case	50 nodal loads, 40 loaded members, 8 support movements, $+50°C$ on all members

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Comprehensive Bridge Case: Load Balance and Key Responses

Applied Actions

- 50 nodal loads with total vertical load $= -3525\text{kN}$
- 40 loaded members with $q_{\text{global}} = (0, 0, -120)$ kN/m → total distributed vertical load $= -43200\text{kN}$
- 8 support movements
- Uniform temperature rise: $+50°C$ on all members

Key Results

- Total applied vertical mechanical load -46725kN
- Support reactions sum to 46725kN in z axis
- Maximum total displacement: $3.2791\ \text{m}$ at node 350
- Largest vertical displacement: $u_z = 0.7352m$ at node 181

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Main Takeaways

Conclusion

The validation cases support the correctness of equilibrium, compatibility, thermal strain, temperature load, member load calculation, and release handling.

The hybrid solver can analyze a much larger mixed truss-frame system in one global model.

For the bridge, critical axial forces concentrate near support-adjacent truss members, while large bending moments concentrate in the arch ribs.

Questions?